Motel

Any research repository → **a verified tool every AI assistant can run** · built for the physical sciences

50	303	10	5	3
research tools	golden tests	hackathon projects	AI apps connected	OS: Linux · Mac · Win

1 BUILD ONCE — your coding agent wraps the code, Motel verifies it

FAILS? THE AGENT FIXES IT AND RE-CHECKS

START
A research repo + its paper

usnistgov/fipy	diffusion, phase field
xraypy/xraylarch	X-ray absorption
iRASPA/RASPA3	adsorption
MauricioCafiero/UMADock	hackathon project

```
$ motel onboard <github-url> --paper <doi>
```

ONBOARD
Your coding agent builds the tool

Claude Code · Codex · Gemini CLI

+ Motel onboarding skill

- Pin the commit; build an isolated environment
- Wrap the science, not the UI or LLM layer
- Fix upstream bugs as replayable patches
- Typed functions, units, validity limits, citation

`motel.yaml` `api.py` `SKILL.md` `patches/` `tests/data/`

VERIFY · GOLDEN TESTS
Must reproduce known science

TOOL	COMPUTED · REFERENCE
RASPA3	CO_2 in silicalite 1.52 mmol/g · exp. 1.48
Psi4	water dimer −4.9184 kcal/mol · S66 −4.918
FiPy	steel carburizing 25,456 s · Callister 25,400
xraylarch	EXAFS Cu–Cu 2.542 Å · crystal 2.556
Zeo++	MFI pore 6.30 Å · IZA database 6.36

Caught before release: overlapping atoms in UMA-Dock sites, truncated AIMNet2 neighbour lists, a five-compound fallback hazard table in Catalyze.

SHARE
A catalog of 50 verified tools

Quantum	PySCF · Psi4 · GPAW · QE · xTB · CREST
Simulation	LAMMPS · OpenMM · phonopy · ML potentials
Materials	pymatgen · PyXtal · Zeo++ · RASPA3
Continuum	Cantera · pycalphad · FiPy · PyBaMM
Spectra	xraylarch · xraylib · pyOpenMS · NMR
Molecules	RDKit · AiZynthFinder · Vina · UMA-Dock

established 36 · published 4 · experimental 10

```
$ motel add zeopp raspa3
```

INSTALLED ON DEMAND

2 USE EVERYWHERE — any AI assistant runs real science through Motel

ASK
A question, in any AI app

> Can silicalite (MFI zeolite) capture CO_2 from power-plant flue gas, 15% CO_2 in N_2, at 1 bar and 300 K?

Claude Desktop · Claude Code · Codex · Gemini CLI · Cursor · dashboard · CLI

Connected through the Model Context Protocol and Agent Skills.

RUN
Motel picks, runs, and records

- **Find** — search ranks established, tested tools first
- **Check** — typed inputs, units, stated validity limits
- **Run** — in the tool's own pinned environment, locally
- **Record** — version, source commit, inputs; repeats come from history

no dependency conflicts · reproducible · data stays local · Linux · macOS · Windows

COMPOSE
The agent chains verified tools

`zeopp.pore_geometry(MFI.cif)`
cavities **6.3 Å**, windows **4.4–4.6 Å**, 3D channels

`raspa3.adsorption_isotherm(MFI, CO₂, 300 K)`
pure CO_2 at 1 bar: **1.52 mmol/g** (measured 1.48)

`raspa3.mixture_adsorption(15% CO₂, 85% N₂)`
CO_2/N_2 selectivity **18.5**

ANSWER
An answer you can check and cite

Yes, selectively. Both gases fit MFI's 4.4–4.6 Å channels, but from flue gas it adsorbs CO_2 **18× over N_2**. Pure CO_2 loads **1.5 mmol/g** at 1 bar.

Trust ESTABLISHED · golden-tested against NIST ISODB and IZA
Cites Zeo++ (Willems 2012) · RASPA3 (Ran 2024)
Limits rigid framework, generic force field
Record `raspa3 0.1.0 · pinned source commit`

IMPACT
Research code anyone can use by asking

Students and researchers run real simulation and analysis codes from the AI assistant they already use: no installs, no environment conflicts, no rewrites. 10 hackathon projects already reusable.

INNOVATION
Agents build the tools, Motel checks them

Turning a repository into a tool becomes agent work, made trustworthy by pinned sources, saved patches, isolated environments, and golden tests.

NOVELTY
A tool layer that grows and proves itself

Not a fixed toolkit: any repository can join, but only after it reproduces reference results, and agents see each tool's maturity and evidence when choosing.

PHYSICAL SCIENCES & MATH
Real codes, checked against real data

Quantum chemistry, molecular dynamics, phonons, adsorption, X-ray spectroscopy, batteries, CALPHAD, PDE solvers; checked against NIST, IZA, S66, textbooks, and exact solutions.